

Amulet Motion Controller Fabrication Document

Layer Stack Legend

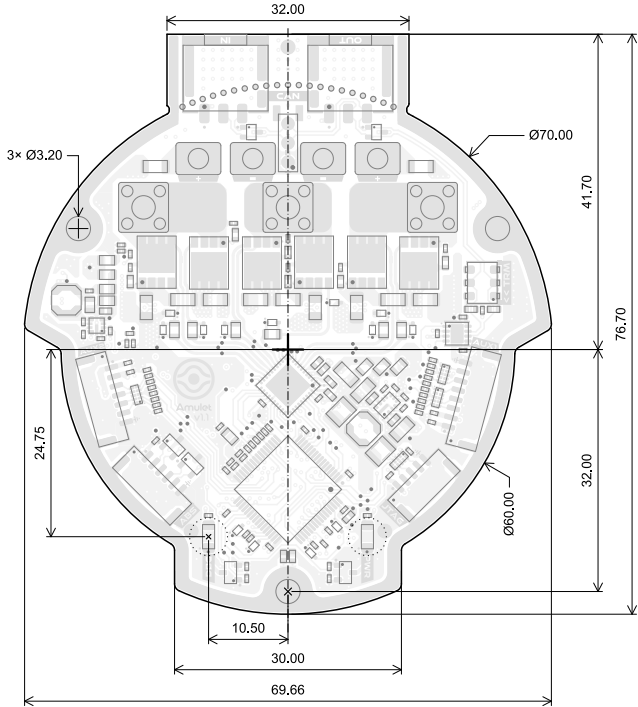
Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
Core		0,4mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Core		0,4mm	FR4	Dielectric	
Copper	L5 (GND)	0,035mm (1,00oz)		Plane	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L6 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY EPFL XPLORE RESEARCH.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:
 - BOARD SIZE 69.666 × 76.700 mm
 - BOARD THICKNESS 1.660 mm
 - TRACE WIDTH 0.200 mm
 - TRACE TO TRACE 0.200 mm
 - MIN. HOLE (PTH) 0.250 mm
 - MIN. HOLE (NPTH) 0.700 mm
 - ANNULAR RING 0.150 mm
 - COPPER TO HOLE 0.254 mm
 - COPPER TO EDGE 0.250 mm
 - HOLE TO HOLE 0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: EPFL Xplore Research		Variant: CHECKED	Git Hash: 4dc9775
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 12